Data Science & AI

Facultat de Matemàtiques i Informàtica

Introducción

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I am a Full Professor at the Mathematics & Computer Science Department, **Universitat de Barcelona,** where I teach Machine Learning and Data Science. I am the Director of the **Master in Fundamental Principles of Data Science**.

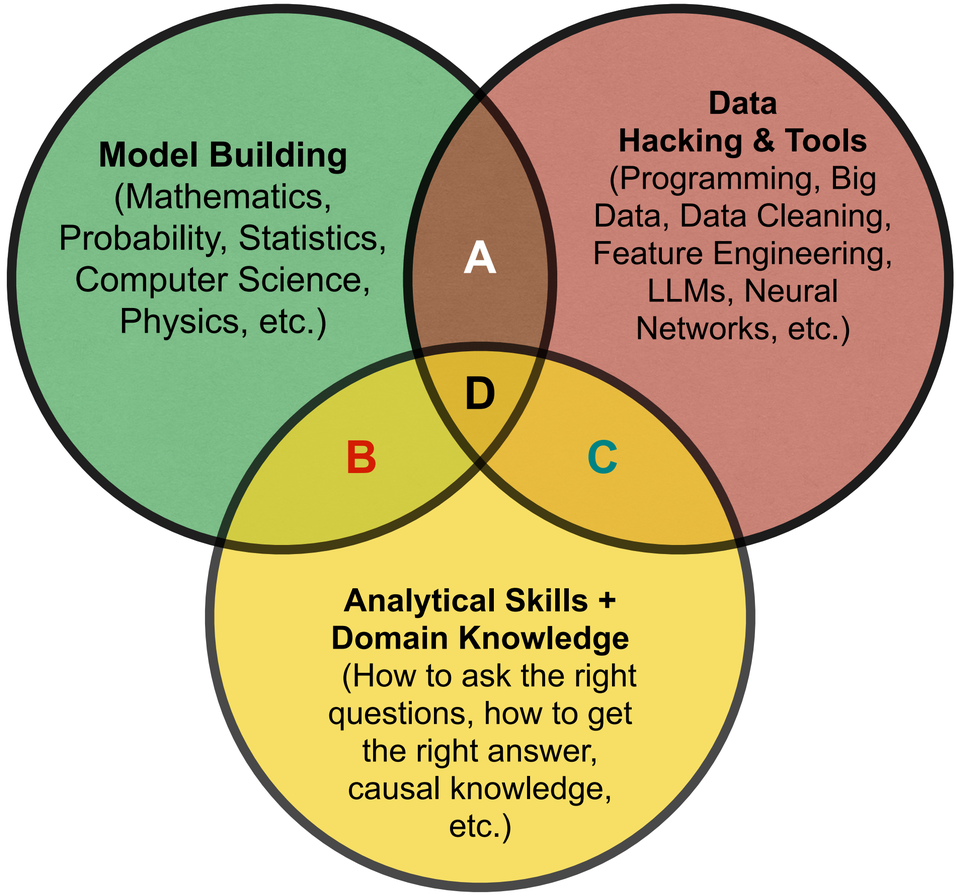
Universitat de Barcelona

- 165 QS World university ranking
- More than 52,169 students
- More than 78 EHEA bachelor's degree courses
- More than 178 master programs
- More than 48 doctoral programs
- More than 6,642 professors, lecturers and researchers

Facultat de Matemàtiques i Informàtica

- Our Faculty is located in the Historic Building, located in one of the most central points of Barcelona.
- The Faculty is home to more than 1,200 students and more than 150 professors, lecturers and researchers.
- The teaching offer of the Faculty includes two undergraduate degrees (CS & MATH), the possibility of taking several simultaneous study itineraries (MATH&CS, MATH&PHYSICS, MATH&BA), and 2+5 university master’s degrees, together with a postgraduate and a doctorate programs.

What is Data Science?



Data Science is a methodology, not a technology.

Its core lies in how we formulate questions, design analyses, interpret evidence, and make decisions from data.

Today, machine learning and AI are powerful tools within that methodology and, increasingly, the best allies of data scientists.

Data Jobs

FP

Data Analyst

Data analysts are responsible for different tasks such as visualizing, transforming and manipulating the data. Sometimes they're also responsible for web analytics tracking and A/B testing analysis.

BSc + Postgraduate Course or MSc + PhD

Data Scientist

Data scientists work closely with business stakeholders to understand their goals and determine how data can be used to achieve those goals. They design data modeling processes, create algorithms and predictive models to extract the data the business needs, and help analyze the data and share insights with peers.

Data Jobs

BSc + MSc

Data Architect

Data architects are IT professionals who are tasked with defining policies, procedures and technologies that will be used to collect, organize, store, and retrieve information for the organization.

BSc + MSc + PhD

ML/AI Engineer

The role of ML/AI engineers involves designing, building, and deploying machine learning systems, as well as monitoring the different systems' performance and functionality.

UNIVERSITAT DE BARCELONA

Tentative Calendar

45 sesiones · **5** miniproyectos · **1** proyecto final

BLOQUE 1 — Fundamentos y datos
15 oct – 29 oct

- 0 — Introducción + ciclo de vida de un proyecto de IA — 15 oct
- 1 — Setup (Python, uv, Git) + AI-Powered Workflow — 20 oct
- 2 — SQL, pandas/polars, calidad de datos, EDA — 22 oct
- 3 — **Miniproyecto: de CSV a dashboard (I)** — 27 oct
- 4 — **Miniproyecto: de CSV a dashboard (II)** — 29 oct

BLOQUE 2 — Estadística y causalidad
3 nov – 26 nov

- 5 — Stats I: distribuciones e incertidumbre — 3 nov
- 6 — Stats II: inferencia y A/B testing — 5 nov
- 7 — Regresión I — 10 nov
- 8 — Regresión II — 12 nov
- 9 — Inferencia causal — 17 nov
- 10 — **Miniproyecto DS end-to-end (I)** — 19 nov
- 11 — **Miniproyecto DS end-to-end (II)** — 24 nov
- 12 — **Miniproyecto DS end-to-end (III)** — 26 nov

BLOQUE 3 — Machine Learning
1 dic – 21 ene

- 13 — Supervised Learning I — 1 dic
- 14 — Supervised Learning II — 3 dic

Festivo 8 dic

- 15 — Supervised Learning III — 10 dic
- 16 — Supervised Learning IV — 15 dic
- 17 — MLflow — 17 dic

Navidades

- 18 — Unsupervised Learning I — 12 ene
- 19 — Unsupervised Learning II — 14 ene
- 20 — **Miniproyecto Clasificación (I)** — 19 ene
- 21 — **Miniproyecto Clasificación (II)** — 21 ene

BLOQUE 4 — Deep Learning
26 ene – 18 feb

- 22 — Deep Learning: Basics — 26 ene
- 23 — RNN + CNN — 28 ene
- 24 — Computer Vision — 2 feb
- 25 — NLP — 4 feb
- 26 — Recommenders I — 9 feb
- 27 — Recommenders II — 11 feb
- 28 — **Miniproyecto Deep (I)** — 16 feb
- 29 — **Miniproyecto Deep (II)** — 18 feb

BLOQUE 5 — IA Generativa y Agentes
23 feb – 30 mar

- 30 — From LLM to apps — 23 feb
- 31 — Prompting — 25 feb
- 32 — Fine-tuning — 2 mar
- 33 — RAG I — 4 mar
- 34 — RAG II — 9 mar
- 35 — Agentes I — 11 mar
- 36 — Agentes II — 16 mar
- 37 — **Miniproyecto IA (I)** — 18 mar

Pascua

- 38 — **Miniproyecto IA (II)** — 30 mar

BLOQUE 6 — Producción y Responsible AI
1 abr – 20 abr

- 39 — MLOps — 1 abr
- 40 — Cloud I — 6 abr
- 41 — Cloud II — 8 abr
- 42 — Responsible AI — 13 abr
- 43 — Explicabilidad — 15 abr
- 44 — Sesgos: el caso COMPAS — 20 abr

FINAL — Proyecto Final
22 abr – 27 may

- K — **Kick-off** — 22 abr
- S1 — Seguimiento — 29 abr
- S2 — Seguimiento — 6 may
- S3 — Seguimiento — 13 may
- S4 — Seguimiento — 20 may

Demo Day
27 de mayo de 2027

What is data science?

Our Objective

From data to knowledge (value)

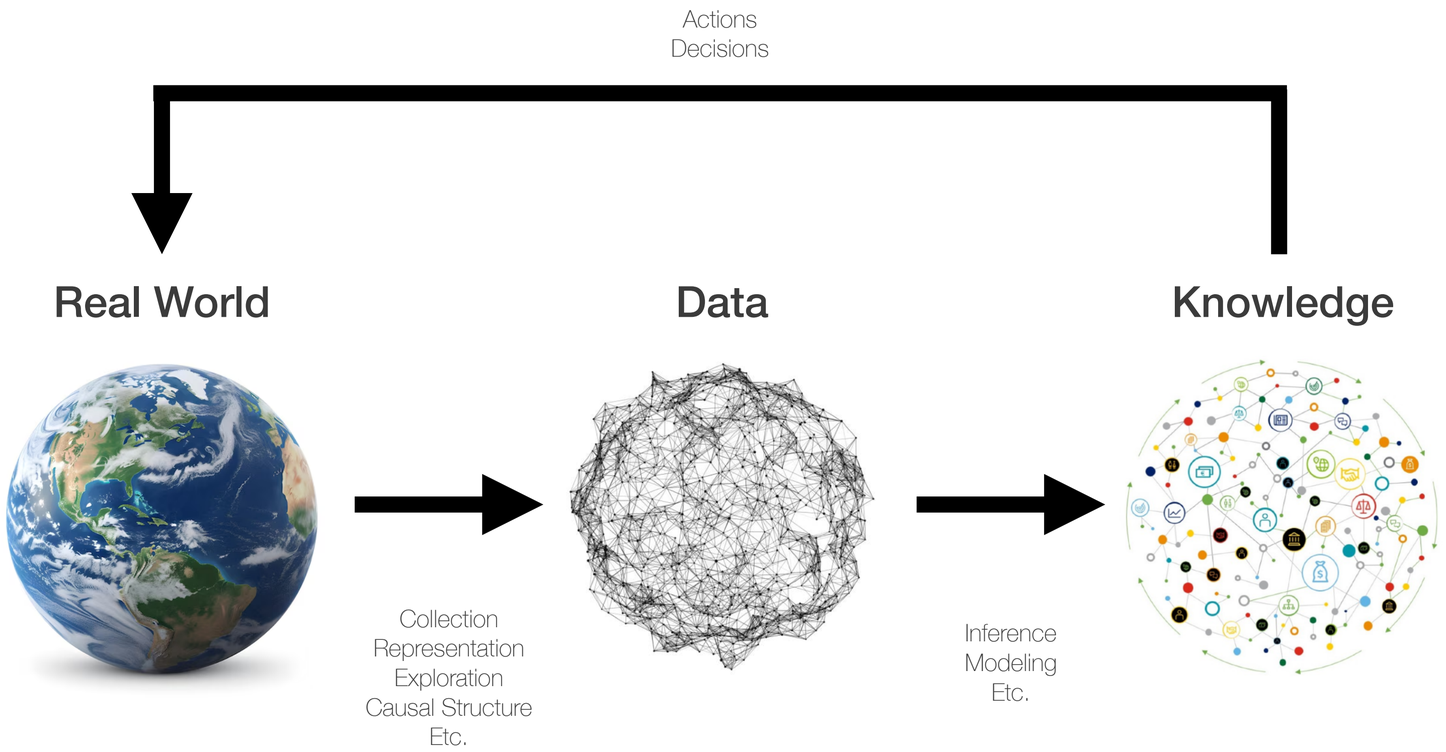
Supermarket dataset

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Total	Date	Time
0	750-67-8428	A	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	548.9715	1/5/2019	13:08
1	226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.8200	80.2200	3/8/2019	10:29
2	631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	46.33	7	16.2155	340.5255	3/3/2019	13:23
3	123-19-1176	A	Yangon	Member	Male	Health and beauty	58.22	8	23.2880	489.0480	1/27/2019	20:33
4	373-73-7910	A	Yangon	Normal	Male	Sports and travel	86.31	7	30.2085	634.3785	2/8/2019	10:37
...	...	...	...	...	...	...	...	...	...	...	...	...
995	233-67-5758	C	Naypyitaw	Normal	Male	Health and beauty	40.35	1	2.0175	42.3675	1/29/2019	13:46
996	303-96-2227	B	Mandalay	Normal	Female	Home and lifestyle	97.38	10	48.6900	1022.4900	3/2/2019	17:16
997	727-02-1313	A	Yangon	Member	Male	Food and beverages	31.84	1	1.5920	33.4320	2/9/2019	13:22
998	347-56-2442	A	Yangon	Normal	Male	Home and lifestyle	65.82	1	3.2910	69.1110	2/22/2019	15:33
999	849-09-3807	A	Yangon	Member	Female	Fashion accessories	88.34	7	30.9190	649.2990	2/18/2019	13:28

Domain knowledge + data science + AI + engineering + data →

- When supermarket shoppers will next visit the store and how much they will spend.
- Which goods are usually purchased together?
- Etc.

From data to knowledge (value)

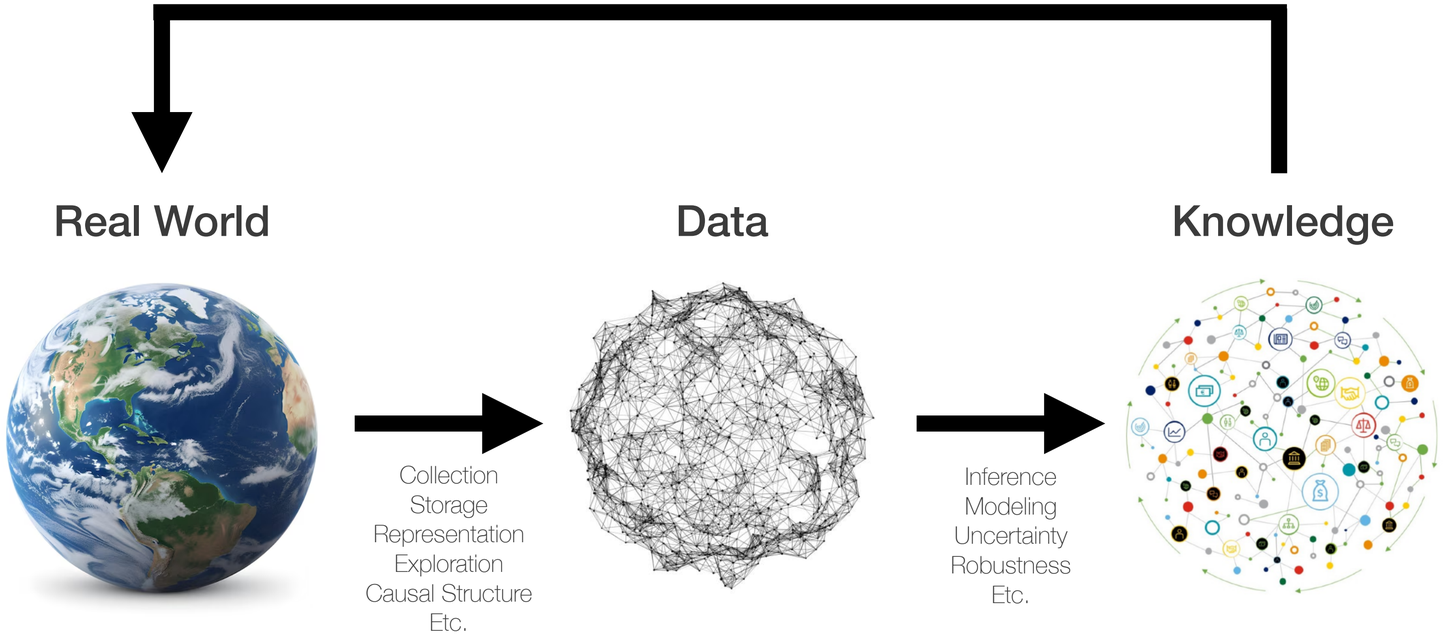


Data (from real world observations and experiments) are the evidence that justify knowledge and actions.

Data Scientists transform data into evidences by using models and knowledge (data can be consistent with a lot different worlds).

From data to knowledge (value)

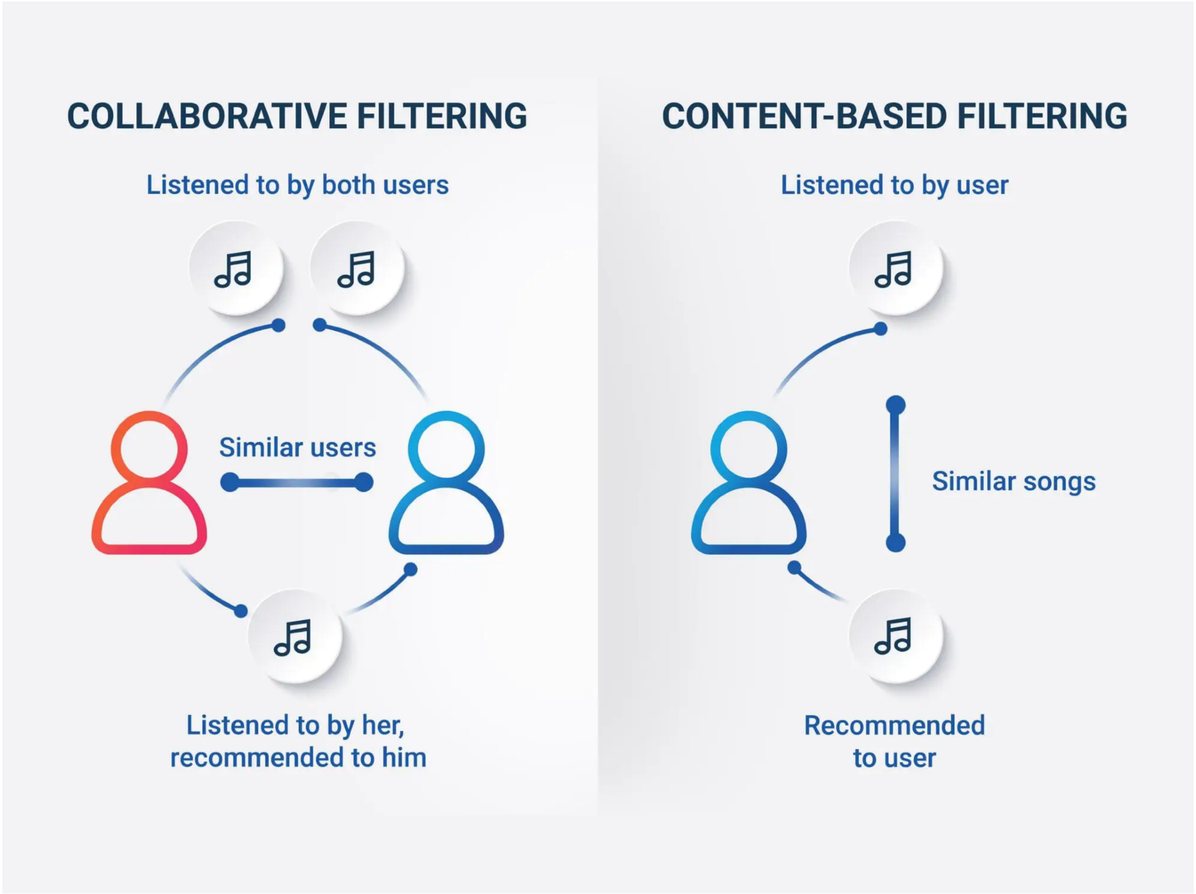
Optimal Actions & Decisions,
Risk Mitigation
Human agency



AI plays a significant role in data science by enhancing the processes of data collection (vision, language, audio), analysis (planning, reasoning), and interpretation (causal models, common sense, etc.).

From data to knowledge (value)

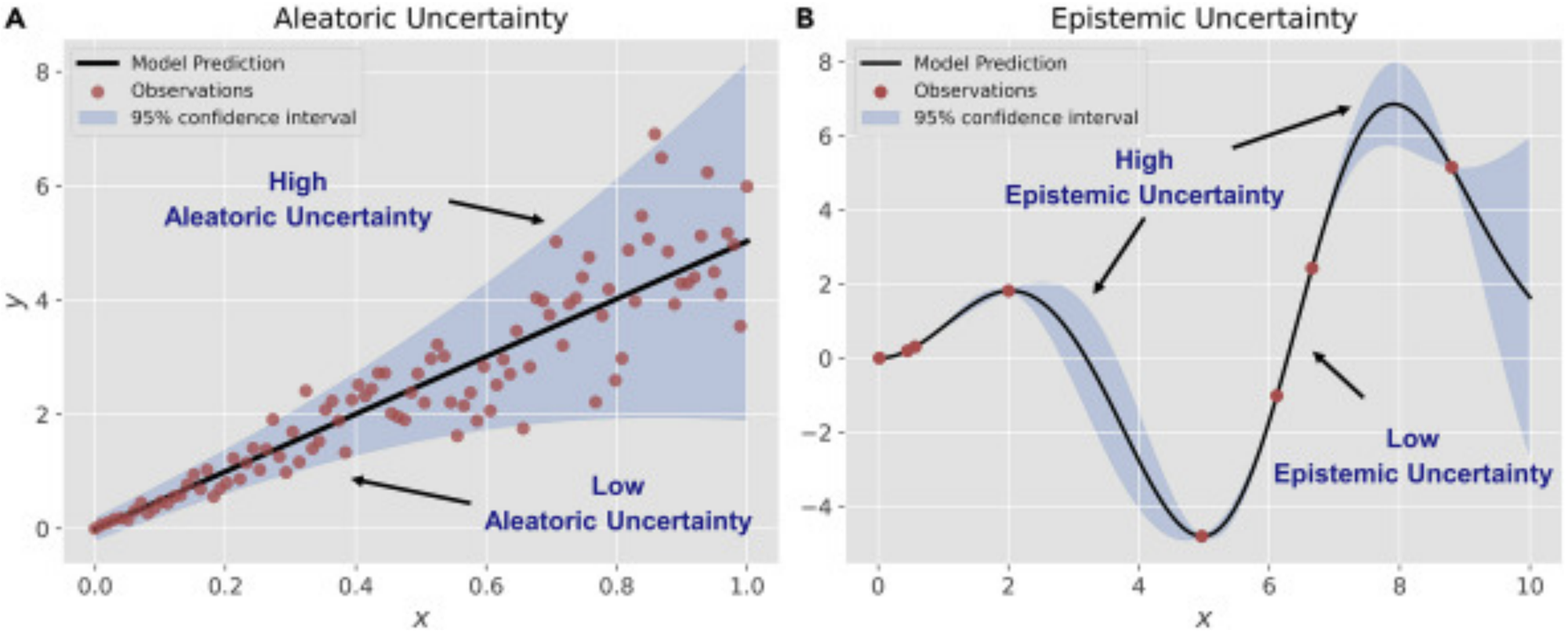
Why is it difficult?



What does it mean to “recommend”?

From data to knowledge (value)

Why is it difficult?

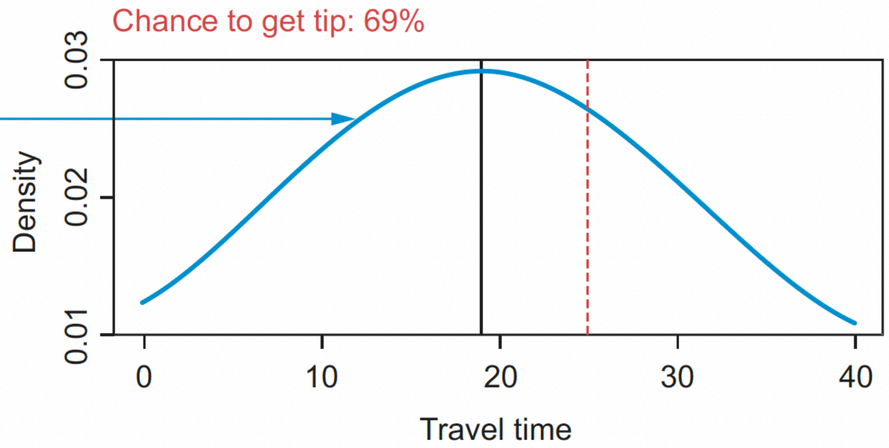


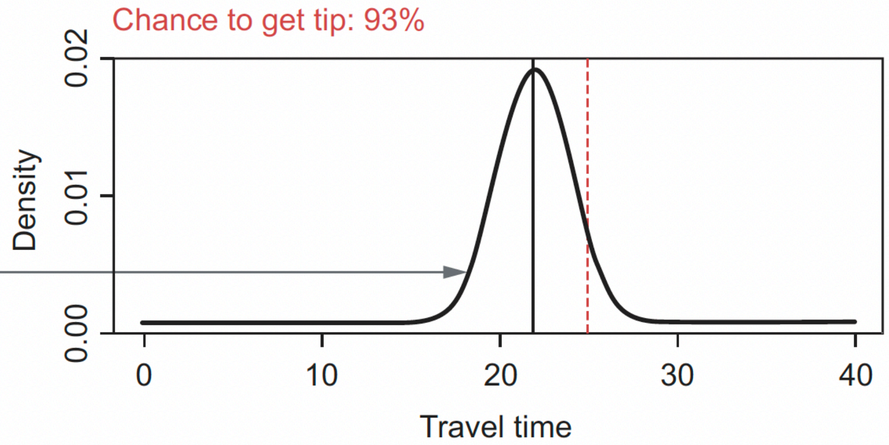
Data produces uncertain evidences

From data to knowledge (value)

Why is it difficult?







Taking the optimal action can be complex

From data to knowledge (value)

Why is it difficult?

	Applicants	Admitted
Men	8442	44%
Women	4321	35%

Table 1: Data From Six Largest Departments of 1973 Berkeley Discrimination Case

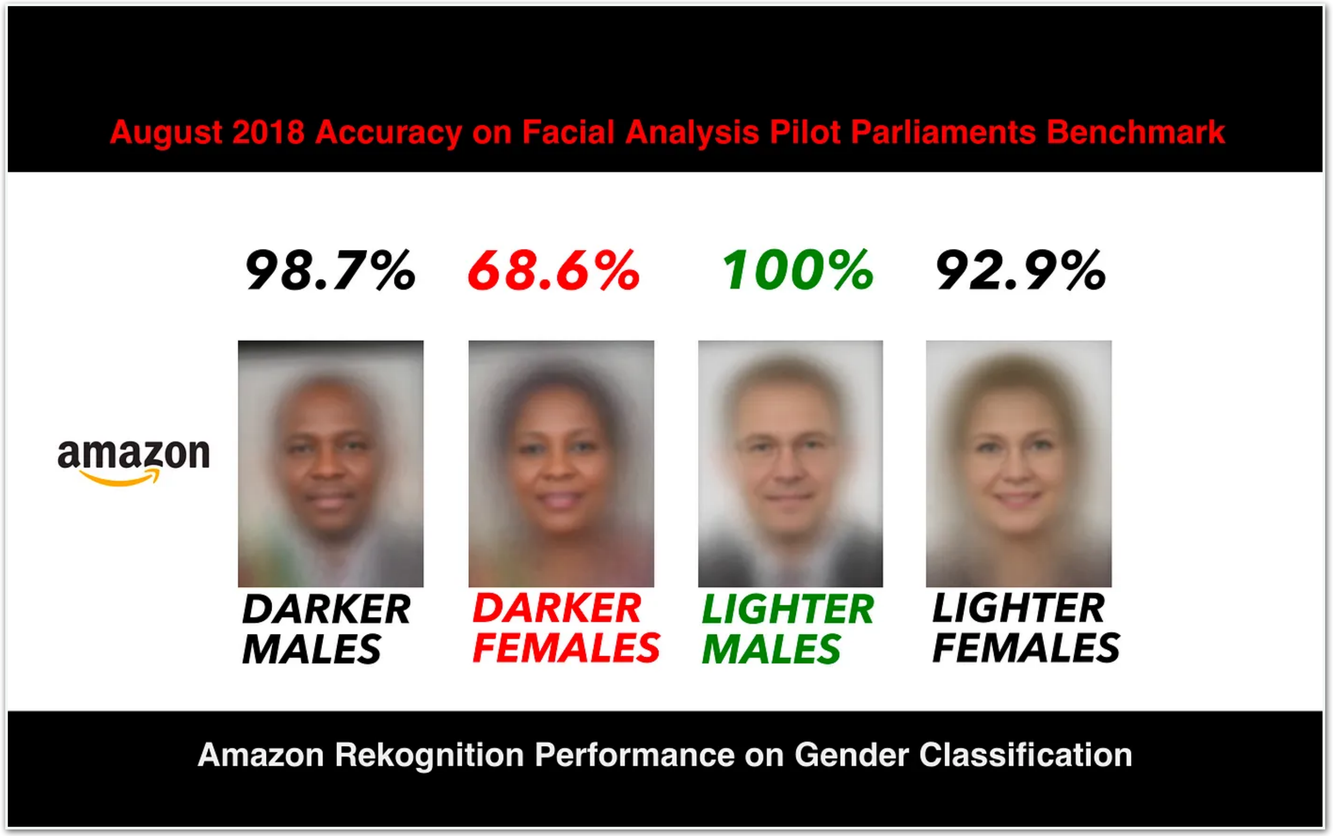
Department	Men		Women	
	Applicants	Admitted	Applicants	Admitted
A	825	62%	108	82%
B	560	63%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	272	6%	341	7%

Source: Bickel, Hammel, and O'Connell (1975); table accessed via Wikipedia at https://en.wikipedia.org/wiki/Simpson%27s_paradox.

Data paradoxes (Simpson's)

From data to knowledge (value)

Why is it difficult?



Data is biased

From data to knowledge (value)

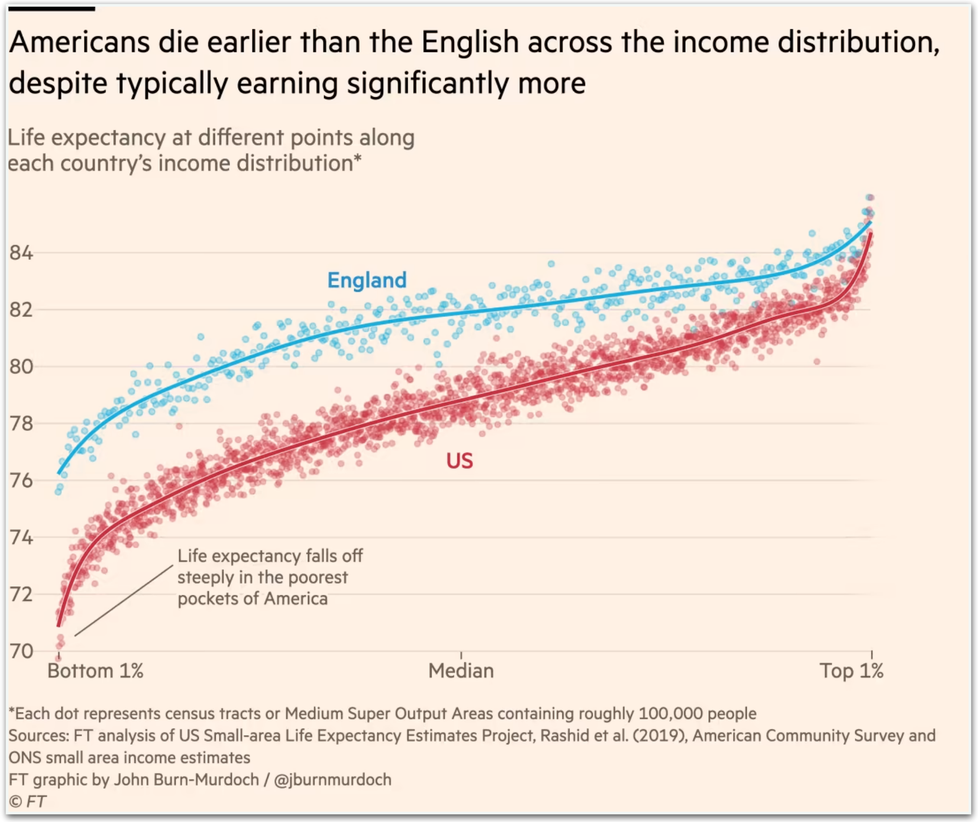
Why is it difficult?



Sometimes, data is scarce

From data to knowledge (value)

Why is it difficult?

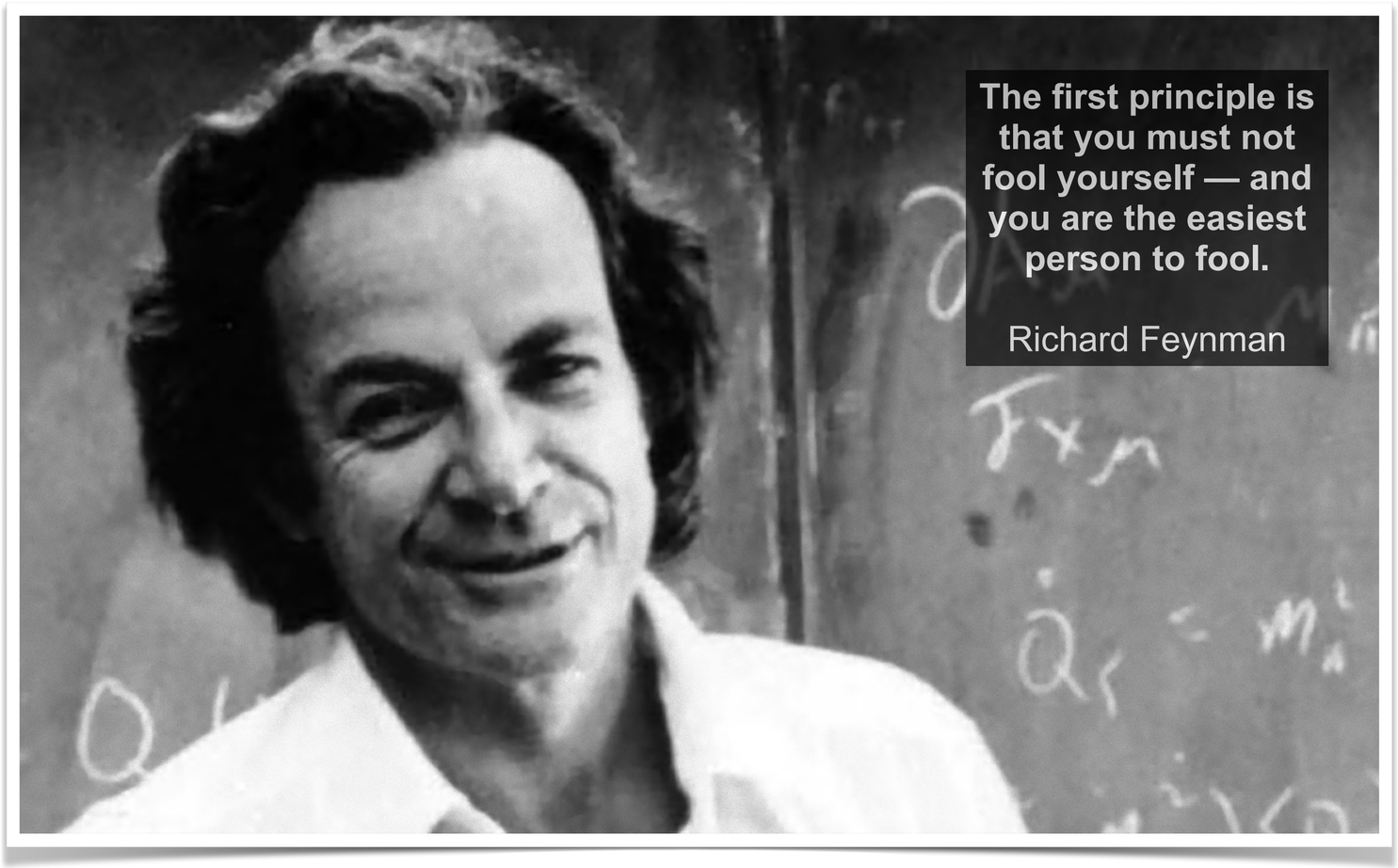




Interpretation is complex

From data to knowledge (value)

Why is it difficult?



Cognitive Biases

From data to knowledge (value)

Why is it difficult?



Following your intuition can be misleading

From data to knowledge (value)

Why is it difficult?

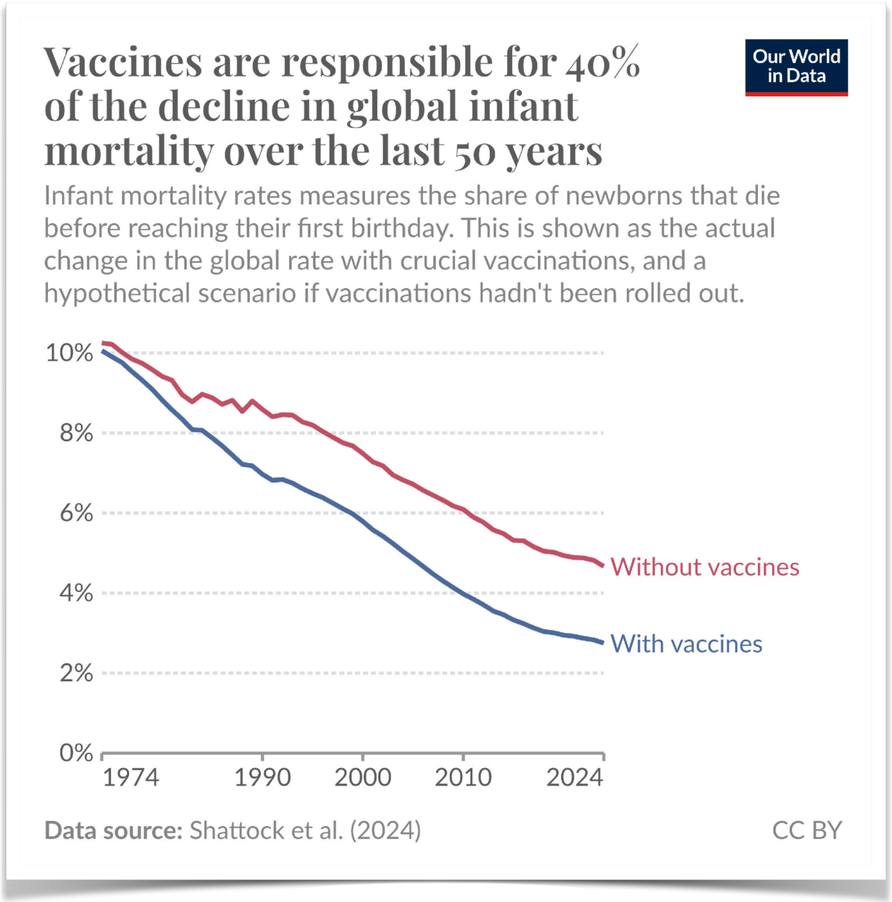


Sentence length and word sophistication have been found to correlate well with the scores of human graders, but they cannot be the base of a program for grading student essays…..

Data is a skewed mirror of reality

From data to knowledge (value)

Why is it difficult?



Observational data methods have a limited “prediction” value. Causality is difficult.

From data to knowledge (value)

Why is it difficult?

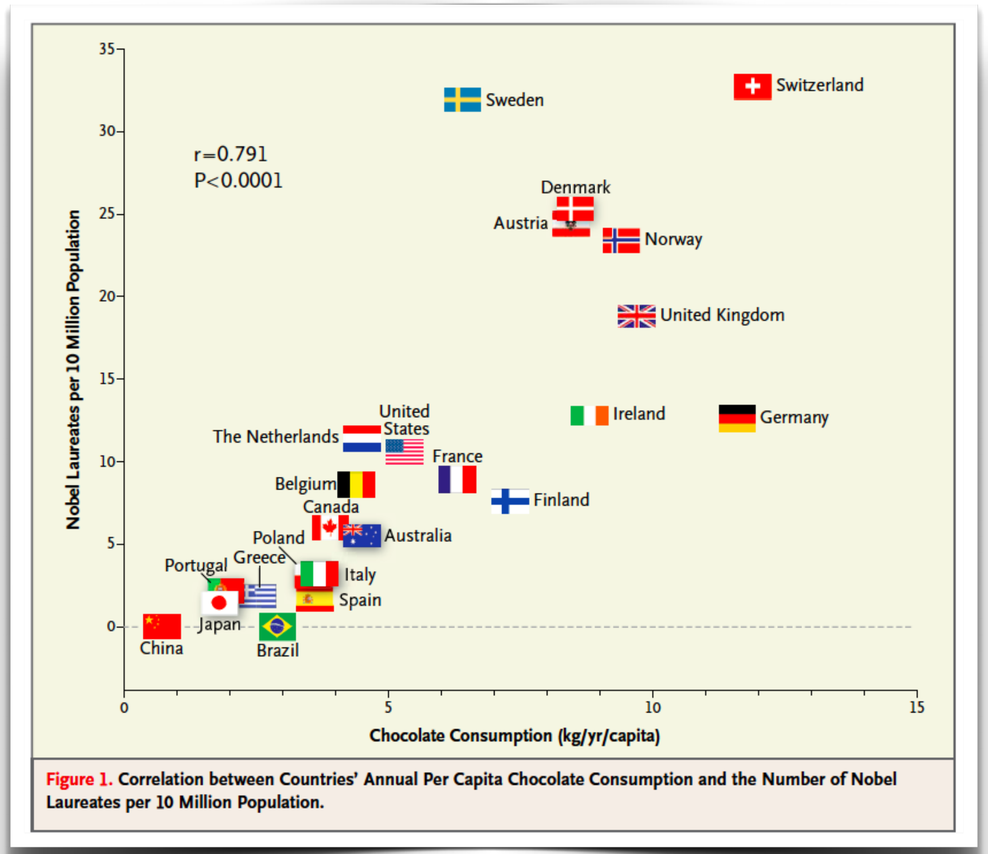
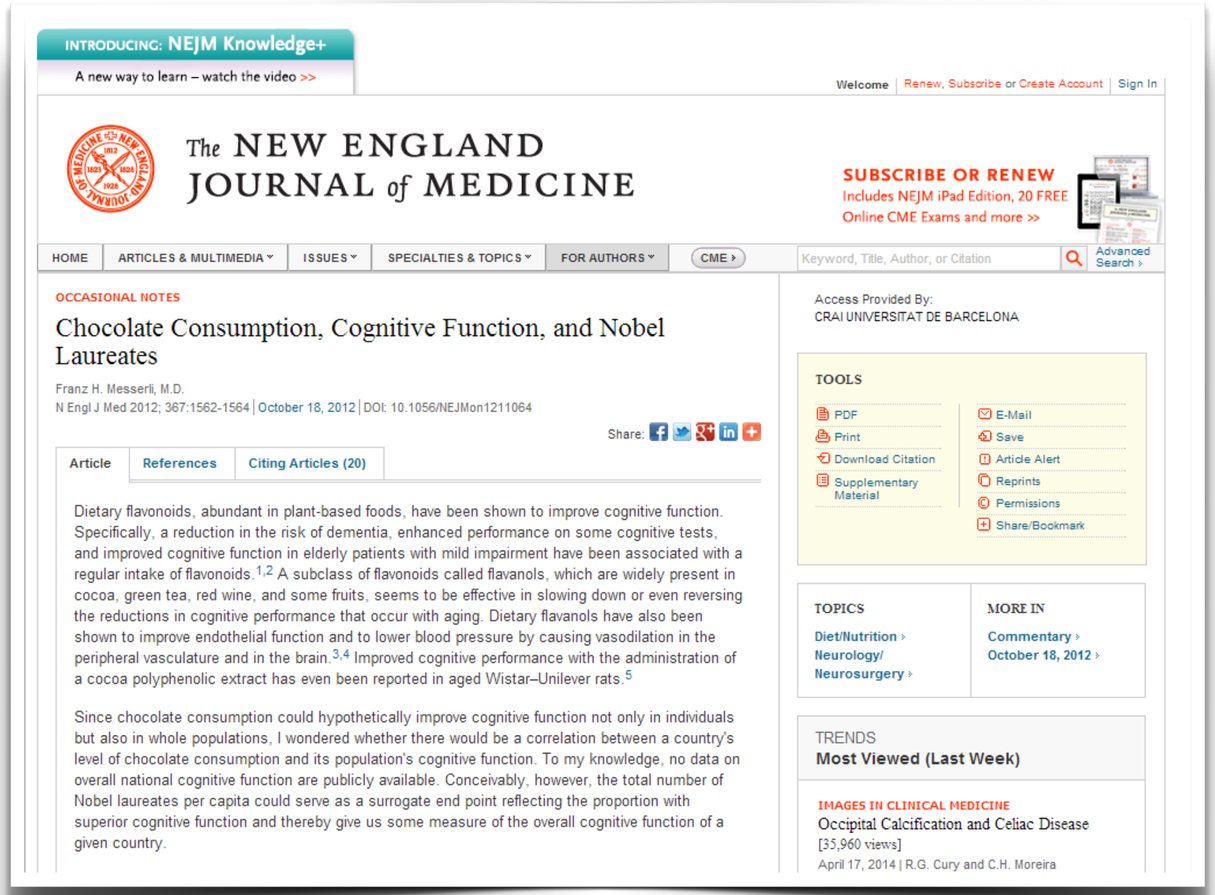


Figure 1. Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.

Bias of automation

From data to knowledge (value)

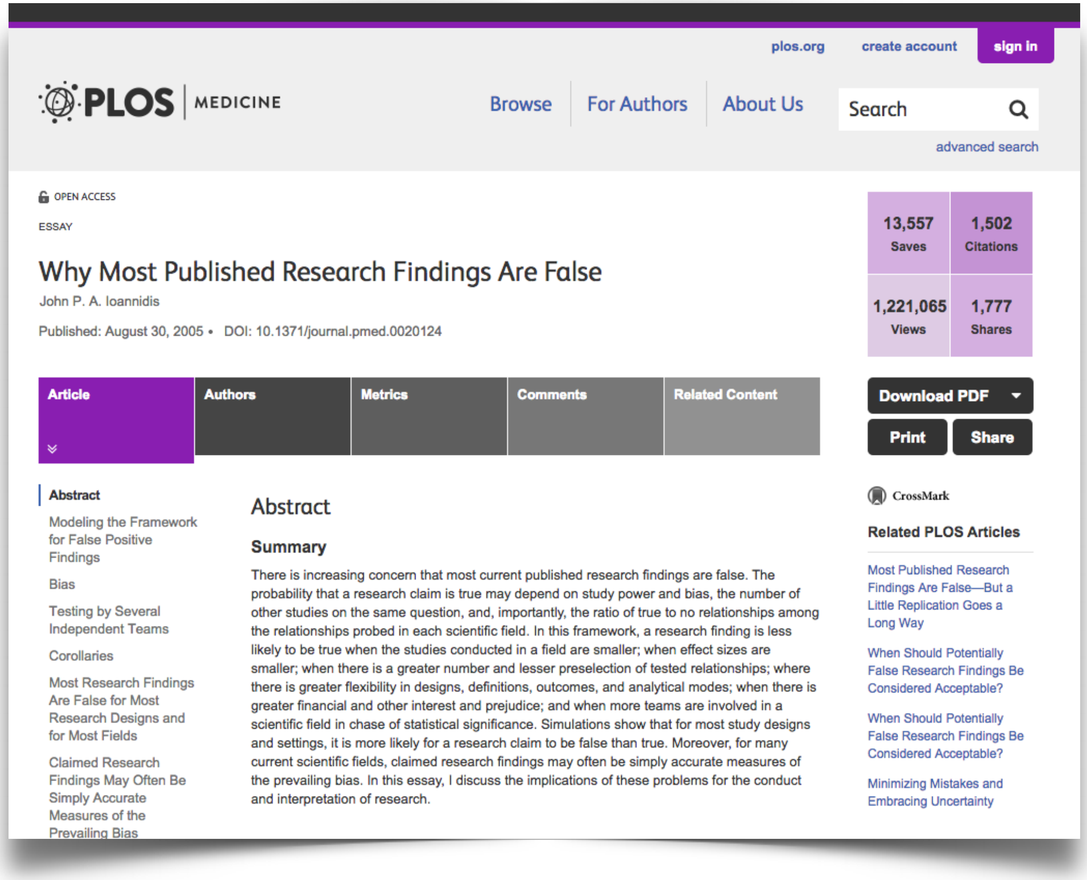
Why is it difficult?



Bias of automation

From data to knowledge (value)

Why is it difficult?



Bias of automation

From data to knowledge (value)

Why is it difficult?



Ethical Issues

From data to knowledge (value)

Why is it difficult?

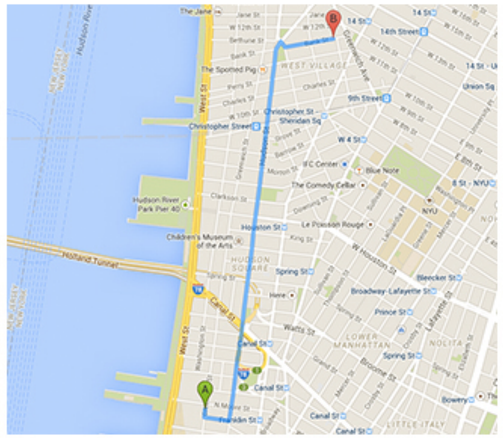
Public NYC Taxicab Database Lets You See How Celebrities Tip

J.K. Trotter
Filed to: DATA 10/23/14 1:00pm

134,190 18



BRADLEY COOPER



JULY 8, 2013 • 7:34 PM - 7:44 PM
376 GREENWICH ST. TO 13 BANK ST.
$9.00 FARE • CASH; UNKNOWN TIP • ©SPLASH

Ethical Issues

Raw Data Sources

1. Acquire — What data do I need, and how do I obtain it?
Identify relevant data sources, assess accessibility and quality, and collect the data needed to address the problem.

2. Process — How do I make the data usable and reliable?
Clean, integrate, transform, normalize and document the data, dealing explicitly with missing values, errors, duplicates and inconsistencies.

3. Enrich — What additional information could improve the analysis?
Create meaningful features, combine complementary sources and incorporate relevant domain knowledge.

4. Summarize — What does the data tell me at first sight?
Describe distributions, patterns, variation and relationships using statistics and visual exploration.

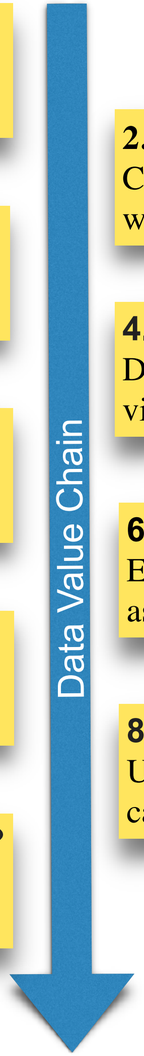
5. Analyze — How can I explain the structure of the data?
Formulate hypotheses, identify relevant variables, model relationships and quantify uncertainty.

6. Discover — What patterns or relationships did I not anticipate?
Explore the data to uncover groups, anomalies, latent structures and unexpected associations.

7. Evaluate — How trustworthy and useful are my conclusions?
Test robustness, validate assumptions, assess uncertainty and evaluate whether results generalize beyond the observed data.

8. Predict — What is likely to happen next?
Use statistical and machine-learning models to estimate future outcomes, classify cases or generate recommendations.

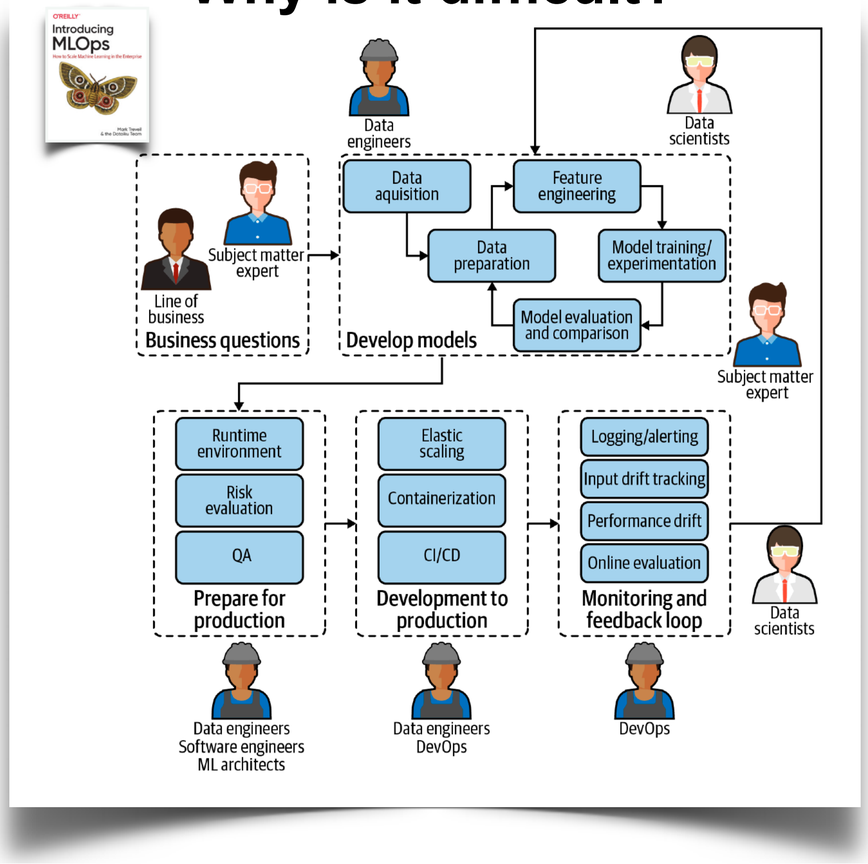
9. Act — What should we do, and what would happen if we intervened?
Use causal reasoning, counterfactual analysis and decision criteria to support actions and interventions.



Insights

From data to knowledge (value)

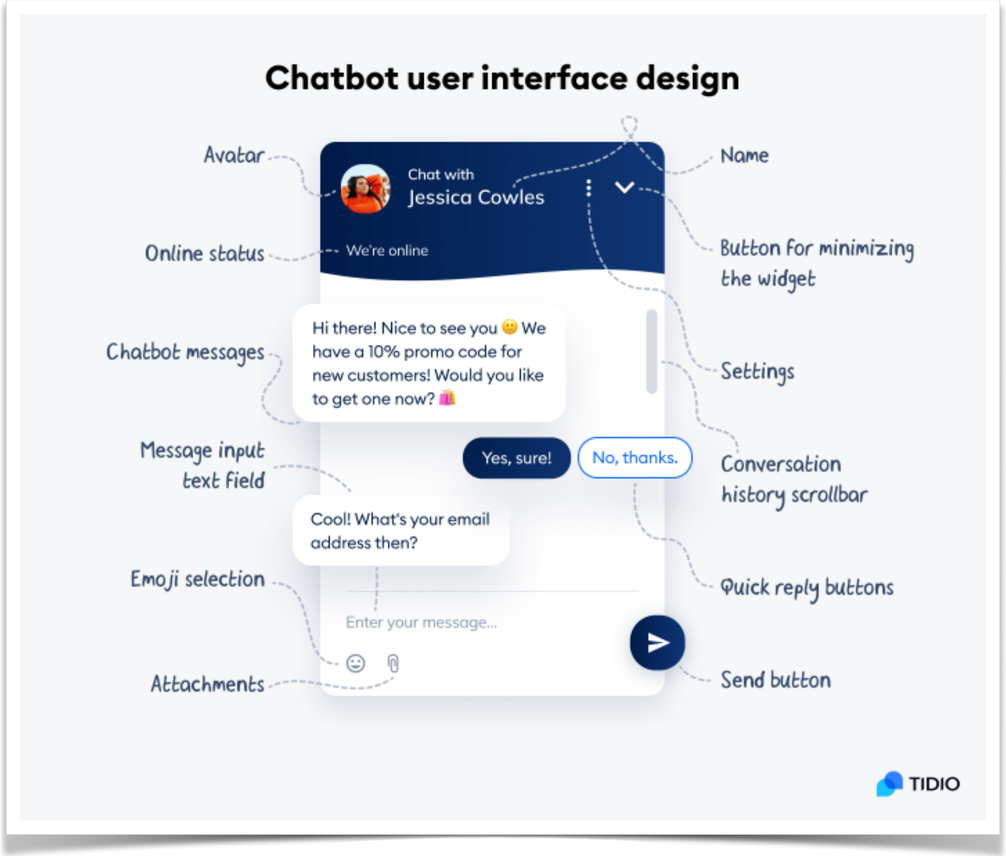
Why is it difficult?

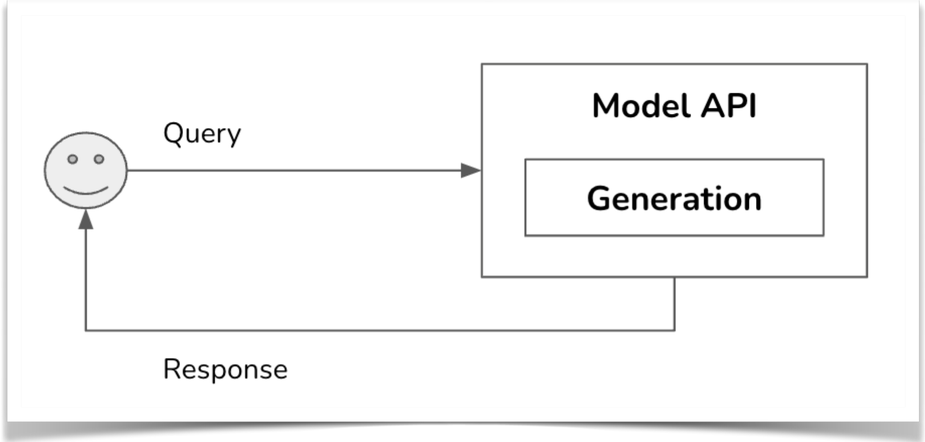


Obtaining and using high quality data and maximinzing its information is difficult

From data to knowledge (value)

Why is it difficult?

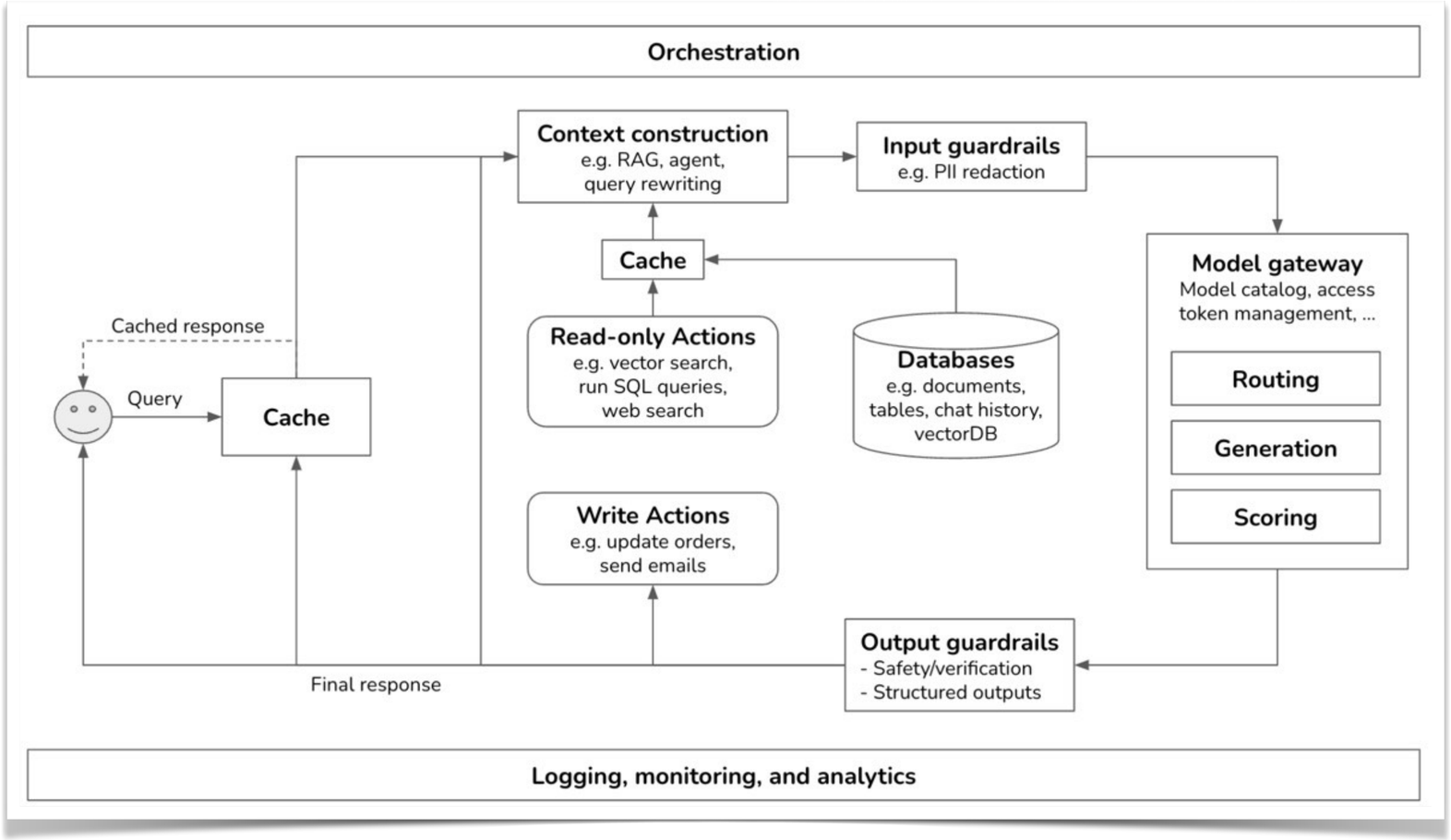




Building good tools is hard

From data to knowledge (value)

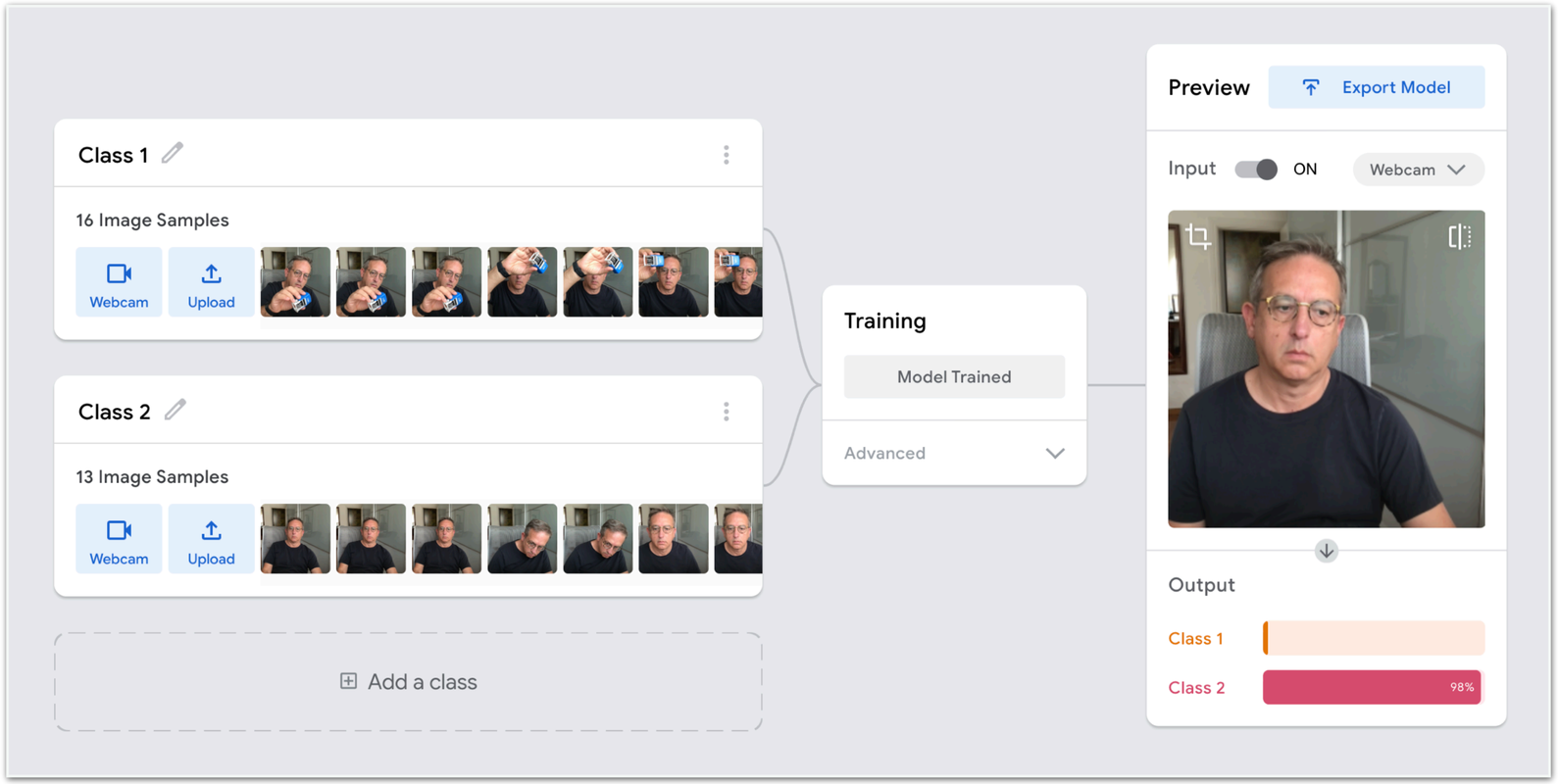
Why is it difficult?



Building good tools is hard

From data to knowledge (value)

Why is it difficult?



https://teachablemachine.withgoogle.com/

What are the limits of Data Science?

Data cannot determine what matters.
Goals, priorities and values come from people and institutions.

Patterns do not automatically explain causes.
Prediction is not the same as understanding or intervention.

Optimization always depends on a choice of objective.
What counts as “better” is a human decision.

Data science cannot replace domain knowledge, judgment or creativity.
Evidence can inform decisions, but it does not eliminate interpretation.

Not everything that can be measured should be optimized.
Ethical and social constraints define the boundaries of legitimate analysis.

> If I had asked people what they wanted,
> they would have said faster horses.
> Henry Ford.

Rethinking data science in the age of AI

YOU

What does data science look like in the age of AI?

ANSWER

Data science in the age of AI is less about doing every step manually and more about deciding what **to delegate**, what **to verify**, and what **still requires human judgment**.

AI can help write code, clean data, build models, explain results, and generate reports. But many analytical decisions depend on **human preferences, domain knowledge, intended uses, values, and context**: what problem matters, which variables are meaningful, what counts as an acceptable error, and how results should be interpreted or acted upon.

So the role of the data scientist is **shifting from producer of analyses to designer, supervisor, and critic of AI-assisted analytical processes**.

Data Science Has Changed

From “write code” to “orchestrate a data workflow”

- Data science is no longer mainly about programming everything yourself.
- Today we combine:
  - data
  - functions, libraries, frameworks and APIs
  - notebooks
  - AI assistants
  - domain knowledge
  - human judgement
- The key skill is not producing code. It is producing reliable knowledge from data.

Modern Data Science Workspace

Notebook + IDE + AI

Typical environment:

- **Jupyter notebooks** for exploration and explanation
- **VS Code** or another IDE for software development
- **Git/GitHub** for version control
- **Python** ecosystem: pandas, NumPy, matplotlib, scikit-learn, PyTorch…
- **AI assistants** integrated into the workflow

A notebook is increasingly a **conversation between data, code, results and reasoning**.

Start With the Question, Not the Model

Before touching the data, define:

- What problem are we trying to solve?
- What decision will depend on the analysis?
- What is the unit of analysis?
- What is the target?
- What information would actually be available at prediction time?
- How will success be measured?

AI can generate 20 models in seconds.

It cannot decide whether you are solving the **right problem in the right way**.

Explore the Data With AI, But Look Yourself

AI can accelerate exploratory data analysis:

- “Describe this dataset and identify potential problems.”
- “Generate code to inspect missing values and outliers.”
- “Suggest useful visualisations.”

But always inspect:

- distributions
- missing values
- duplicates
- strange categories
- impossible values
- correlations
- temporal patterns

Rule: Never outsource looking at the data.

Use AI as a Coding Assistant and Knowledge Partner

AI is excellent for:

- generating boilerplate code
- explaining unfamiliar libraries
- translating an idea into Python
- debugging
- refactoring
- writing visualisation code
- suggesting alternative implementations
- etc.

Specify → generate → understand → test → adapt → use

Work Iteratively in Notebooks

A good notebook is not a pile of cells.

Structure it as an argument:
Question → Data → Exploration → Method → Validation → Interpretation

Useful practice:

1. Small executable steps
2. Markdown explaining why
3. Visualise intermediate results
4. Re-run from the beginning
5. Remove dead experiments
6. Make assumptions explicit

The notebook should tell the story of the analysis.

AI Can Suggest Models; You Must Design the Experiment

AI can propose:

- regression
- trees
- random forests
- boosting
- etc.

The data scientist must decide:

- train/test split
- cross-validation
- baseline
- evaluation metric
- what comparison is scientifically meaningful

A sophisticated model with a bad experiment is still bad data science.

Verification Becomes a Core Skill

With generative AI, **producing an answer is cheap**. Verifying it becomes **valuable**. It is the main bottleneck of the process.

Check:

- Does the code actually do what it claims?
- Are train and test data separated correctly?
- Is there data leakage?
- Are metrics relevant and computed correctly?
- Are results reproducible?
- Do conclusions follow from the evidence?

Think like a scientist:

What evidence would convince me that this result is correct?

Delegate, Describe, Discern, Be Diligent

Four useful competencies for **working with AI**:

- **Delegation**
  Decide what to give to AI and what requires human judgement.
- **Description**
  Explain the task, context, constraints and expected output clearly.
- **Discernment**
  Critically evaluate AI-generated code, analyses and conclusions.
- **Diligence**
  Ensure reproducibility, privacy, attribution and responsible use.

AI literacy is becoming part of **data science literacy**.

The New Data Scientist

The competitive advantage is no longer: "*I can write Python faster than you.*"

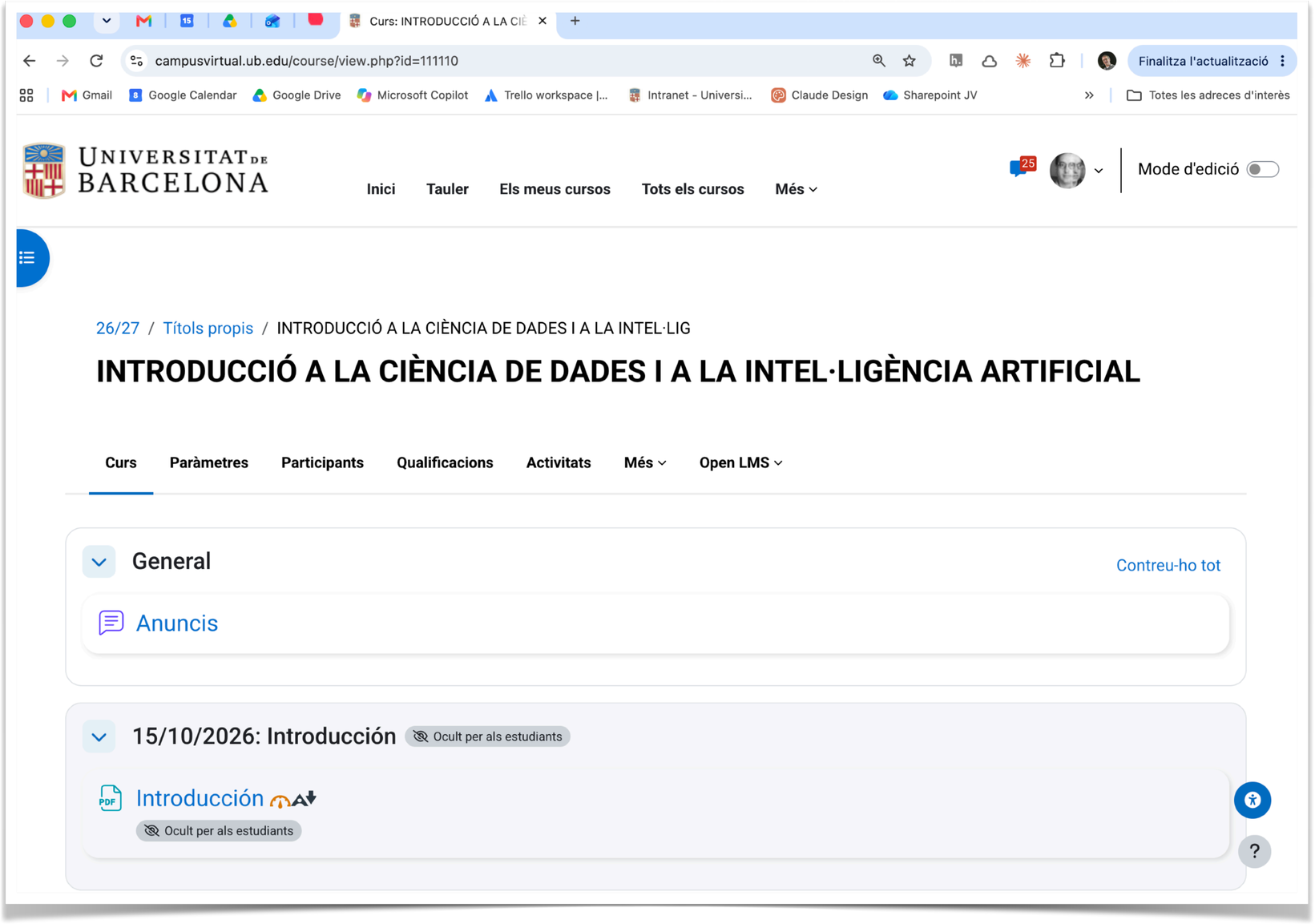
It is: "*I can ask the right questions, leverage AI effectively, interrogate data without torturing it, design rigorous experiments, and recognize when the results can or cannot be trusted.*"

The modern workflow:

Ask → Explore → Delegate → Code → Test → Validate → Interpret → Communicate

AI increases the speed of data science.
Human judgment is now the bottleneck and the main determinant of quality.

Be ready for the next session



Si eres estudiante nuevo de la **Universidad de Barcelona (UB)**, primero debes activar tus credenciales de acceso y después entrar en el Campus Virtual.

1. Obtén tu usuario y contraseña

Si es la primera vez que accedes a los servicios de la UB:

1. Entra en https://auten.ub.edu/auten/NovaPasswordAlum?lang=es
2. Introduce tu **DNI, NIE o pasaporte** y tu fecha de nacimiento.
3. Recibirás un correo electrónico en la dirección personal que utilizaste durante la admisión.
4. Sigue las instrucciones para crear tu contraseña.
5. El sistema te mostrará tu **identificador UB de cuatro caracteres**. Guárdalo porque será tu usuario de acceso. auten.ub.edu

2. Accede al Campus Virtual

1. Abre **https://campusvirtual.ub.edu/**
2. Haz clic en **«Inicia sesión»**, arriba a la derecha.
3. Introduce tu identificador de cuatro caracteres y tu contraseña.
4. Una vez dentro, selecciona **«Mis cursos»** para acceder a tus asignaturas.

A UB student can use **VS Code + GitHub Copilot** through GitHub's student program.
Students can obtain **GitHub Copilot Student for free** after GitHub verifies their student status.

Recommended setup for a UB student

1. **Create a GitHub account**
   - Ideally use or add the **UB institutional email address** to the GitHub account.
   - GitHub may also ask for proof of current enrolment.
2. **Apply for GitHub Education**
   - Go to GitHub Education and request student verification.
   - GitHub accepts a university-issued email and/or dated proof of enrolment.
3. **Activate GitHub Copilot Student**
   - Once verified, go to your GitHub Education benefits.
   - Activate **Copilot Student**.
   - The student plan is free for verified students.
4. **Install Visual Studio Code**
   - Install VS Code on the laptop.
   - Install the extensions:
     - **Python**
     - **Jupyter**
     - **GitHub Copilot**
5. **Sign into GitHub from VS Code**
   - Click the Accounts icon in VS Code.
   - Sign in using the same GitHub account that has the student benefit.
   - Copilot should then become available inside VS Code.

What do you need in your laptop?

1. Visual Studio Code: **Main working environment for the course.**

- Install Visual Studio Code and the Python and Jupyter extensions.
- Install uv, a fast Python package and environment manager.
- Install Copilot in VS Code

2. Git + GitHub: Clone repositories, commit changes, collaborate and keep a history of your work

- GitHub account
- GitHub Desktop
- Git installed locally

Repository

We have created https://github.com/DataScienceUB/posgradoDSAI2026, where we will upload guides and code whenever needed. The README will serve as the index of resources.

What do you need in your laptop?

3. A project folder, not isolated files

https://github.com/DataScienceUB/posgradoDSAI2026

```
my-project/
├── AGENTS.md          instructions for the AI assistant (read it first, always)
├── SPECS.md           WHAT we build
├── DESIGN.md          HOW it should look / behave
├── ARCHITECTURE.md    HOW it is built
├── TASKS.md           the to-do list, in small steps
├── README.md          for people
├── pyproject.toml, uv.lock
├── data/              (.gitkeep keeps the empty folder in Git)
├── notebooks/         exploration
├── src/               reusable code (.py)
└── tests/             tests (pytest)
```

The key idea

Your laptop becomes a **human + AI data science workspace**:

VS Code + GitHub + Copilot + Jupyter + Python + your data

You work in the project; the AI works **with the context of the project**, rather than just answering isolated questions in a chatbot.